# Climb the Ladder: Programming Practice Worksheet

**Intro**

Hey there, in this lesson we are going to get busy applying our new knowledge of PLC programming with some examples.

Note: if you don't have access to Studio 5000 Logix designer or RSLogix 5000 I've added alternate instructions for doing the same thing with the free RSLogix Micro software

Here's the examples we will be working on:

- Liquid Level Control: with computed scaling
- Generator Activation With Load Comparison

**Hardware Setup:**

For both of the problems below set up your controller and I/O as follows:

**Logix5000 controllers (RSLogix 5000/Studio 5000 Logix Designer):**
Set up a ControlLogix system with the following components:

- One 1756-L71 (or 5570 Emulate to test with the Emulator) processor, slot 0
- One 1756-OB16E 24VDC Output module, slot 1
- One 1756-IB16 24VDC Input module, slot 2
- One 1756-IF8 analog input module, slot 3

**Problem 1:**

Open and close a discrete valve to keep a tank within a particular range. This example is very similar to what we did in the previous lessons except for a few things. Instead of a water tank this is a solution tank. But that shouldn't make a difference to the program, right?

Right.

Also, instead of doing the scaling in the module configuration, I'm going to have you do the scaling with programming.

And the last difference is we will be controlling a fill valve that is a discrete valve. In other words, it's either open or closed...no in between values, which will make that part a little bit easier.

<u>So here's the problem:</u>

*The solution tank needs to be actively filled anytime it is less than 80% full, and it will continue filling until it reaches 90%. So it starts filling till it gets to 90% then stops filling until the tank level gets down below 80%.*

Now, you need to be able to work with the tank level analog input in terms of gallons, but it's coming into the PLC as a 0-10V signal .

So you're going to need to use math instructions to convert  0 - 10V to 0 - 480 gallons which is the size of our solution tank.

So I want you to use regular math instructions, not the compute instructions. So to make things a little easier for you the math formula to calculate gallons would be something like this:

$$\left(\frac{Analog\ \ Input\ Value}{Analog\ Input\ Max}\right)\ \times\ Tank\ Level\ Max\ =\ Scaled\ Value$$

So in this case: Analog Input Max is 10 (volts)  and Tank Level MAX is 480 gallons).

So that's how you'll scale the data coming in.

Output point 11 of the OB16E will control the fill valve for the tank. Channel 2 of the IF8 module will be your analog input. Be sure to set up the channel configuration for 0-10V and the scaling should just be set to 0-10V equals 0-10 engineering units because you are going to do the scaling in the logic with math instructions.

I/O should be connected as follows:

- Create a new tag called "**Fill_Valve**" and alias to the output 11 of the 1756-OB16E output module. This will control the fill valve for the tank.
- Create a new tag called "**Tank_Level_Raw**" and alias it to Channel 2 of the IF8 module (Slot 3). This will be your analog input for tank level. Be sure to set up the module channel configuration for 0-10V and the scaling should just be set to 0-10V equals 0-10 engineering unit because you are going to do the scaling in the ladder logic with math instructions.

REMEMBER TO CHANGE CONTROLLER EMULATE 5570 IF YOU WANT TO TEST GOING ONLINE WITH LOGIX EMULATE.

**MicroLogix controllers (RSLogix Micro)**
(If you haven't already be sure to watch the analog instructions and config tutorial from Ladder Logic Foundations if you haven't already):

https://www.myplctraining.com/products/the-confident-plc-programmer-method-rockwell/categories/1986972/posts/7627138

Use a MicroLogix 1100 series B

- Add a 1762-IF4 analog input module

-10 to +10V will automatically be represented by -32760 to +32,760, so 0-10V will be represented by 0-32,760.

You'll still use this formula with math instructions to calculate the scaled value but your input MAX will not be 10 in this case. It will be 32,760

$$\left(\frac{Analog\ \ Input\ Value}{Analog\ Input\ Max}\right) \times Tank\ Level\ Max = Scaled\ Value$$

Use the first available discrete out for the tank fill valve: O:0.1
Use the first available analog input for the tank level signal: I:1.0

**Problem 2:**

Ok, our second example will be related to the power generation industry. This is where most of my PLC experience is, so I really enjoy these type problems.

So here we go:

*We have two 1000kW generators that power a facility on certain days. In most cases one generator can handle at least double the load of the facility. But sometimes in the summer when it's really hot and air conditioning load is high, the second generator is needed to prevent the first generator from being overloaded. There is a lot of logic that could go into this, but we will keep this relatively simple for now. You need to create basic logic that will activate the second generator if the kW load on the first generator has been greater than 60% of the capacity of the first generator for more than 2 minutes.*

For the purpose of this exercise we will only start the generator. We won't worry about closing the breaker. But in REAL LIFE you would need to tell a circuit breaker to close to put the generator online in parallel with the first generator.

First create a start/stop generator rung for generator 2, like you did in Ladder Logic Foundations using the make-seal-breaker method (latch/seal-in logic). Then in addition to the start rung add two other rungs. One that starts timing once the first gen's (Gen 1) load goes over 60%, and a second rung that turns on "Activate_Gen_2" boolean tag (or B3 file in RSLogix Micro). Then add your Activate_Gen_2 bit into the generator start rung such that either the start push button OR the Activate_Gen_2 bit will start generator 2.

I/O should be connected as follows:

- Output 2 of the 1756-OB16E (Local:O: output module will control the Gen 2 start relay.
- Input 1 of the 1756-IB16 should be the Start push-button for Gen 2
- Input 2 of the 1756-IB16 should be the Stop push-button for Gen 2
- Channel 0 of the 1756-IF8 should be the generator kilowatts (kW) signal for Gen 1. Configure for 0 - 10V and scale to 0-1000kW

**MicroLogix controllers (RSLogix Micro)**

Use a MicroLogix 1100 series B

- Add a 1762-IF4 analog input module

-10 to +10V will automatically be represented by -32760 to +32,760, so 0-10V will be represented by 0-32,760. Scale to 0-1000kW for the generator load calculation and use that for your comparison instruction(s).